Chemotherapy-Induced Peripheral Neuropathy (CIPN) and the Use of Hilotherm for Prevention

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Background

- CIPN is a common side effect of taxane-based chemotherapy in breast cancer patients.
- It negatively impacts quality of life and can lead to treatment discontinuation.
- Hilotherm® offers temperature-controlled hand-foot cooling and may reduce CIPN severity.
- This evaluation compares real-world data with Schaper (2023) as a control.

Methods

Participants

- 44 breast cancer patients receiving taxane chemotherapy

Inclusion Criteria

- Breast cancer diagnosis
- Taxane-based regimen
- Hilotherm initiated from cycle 1
- Clinician support and patient consent

Data Collection

- Standardised paper form or Microsoft Teams form at each cycle

Limitations

- No local control group
- Incomplete data for some cycles
- Occasional data entry errors (e.g., hospital number discrepancies)

Results

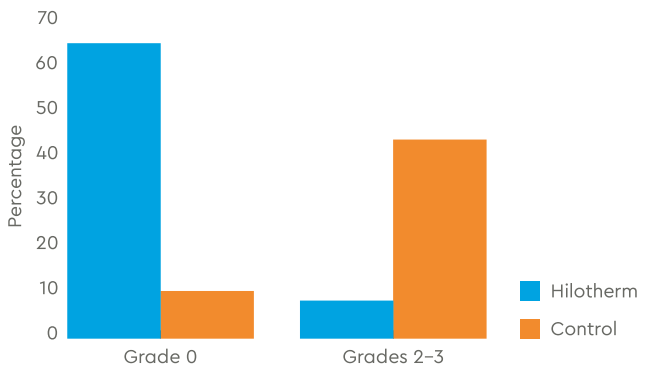
CIPN Incidence

- **Grade 0 (No symptoms):** 63.6% (Hilotherm) vs. 9.5% (Schaper control)
- **Grades 2–3 (Moderate to severe):** 6.8% (Hilotherm) vs. 45.8% (Schaper control)

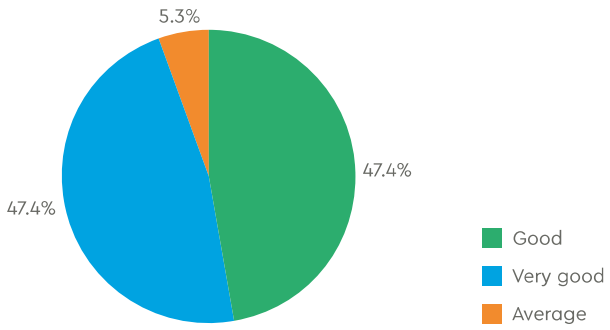
Patient Experience

- "Very good": 45 responses
- "Good": 45 responses
- "Average": 5 responses

CIPN Grades Comparison

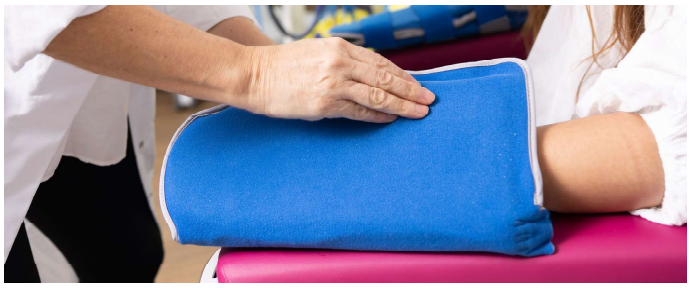


Patient Experience Ratings



Conclusion

- Hilotherm therapy significantly reduced CIPN incidence and severity.
- Patient feedback was overwhelmingly positive.
- Despite limitations, findings support further use and investigation of Hilotherm in other treatment and tumour groups.
- Larger, controlled studies are recommended.



References

1. Schaper, T., Dr (2023). Is the efficiency of temperature-controlled hand-foot- cooling (HILOTHERAPY®) in prevention of CIPN sustainable – update on the 4-year follow-up. German Society for Senology. https://doi.org/42. Jahrestagung DGS 2023

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